

ASSESSMENT TOOL – 3

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO2	Assessment	Debugging & Optimization Task
Weightage		Total Marks	

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Section / Batch	ECE-2024	Date of Submission	29/7/26
Faculty Incharge	Dr R S Selvan	Assessment	Individual Submission

Total Marks	20	Obtained Marks	19
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Code of Conduct:

I Cindy R / 192412571 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

1. IPv4 Network Debugging

Given

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

Step 1: Identify the Subnet

A /26 subnet has a block size of 64.

Subnet ranges are:

Subnet	Address Range
192.168.10.0/26	0-63
192.168.10.64/26	64-127
192.168.10.128/26	128-191
192.168.10.192/26	192-255

Since the IP is 192.168.10.65, it belongs to

Subnet: 192.168.10.64/26

Step 2: Calculate Network Details

Network Address

= 192.168.10.64

Broadcast Address

= 192.168.10.127

Host Range

= 192.168.10.65 – 192.168.10.126

Step 3: Verify IP Address

Configured IP

192.168.10.65

It lies inside the host range.

2 IPv6 Debugging

Given

Device A

2001:db8:ff00:42::8329/64

Device B

2001:db8:0:0:1::1/64

Step 1: Expand Addresses

Device A

2001:db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network Prefix

With /64

First 64 bits are network.

Device A

2001:db8:0000:0000::/64

Device B

2001:db8:0000:0000::/64

Step 3: Same Subnet?

Yes.

Both have identical first four blocks.

✓ Same subnet.

Step 4: Addressing Problem

There is no prefix mismatch.

Possible causes:

- Firewall
- Duplicate Address Detection failure

5. Routing Table Debugging

Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Destination Packet

192.168.2.100

Step 1: Longest Prefix Match

Destination

192.168.2.100

Matches

192.168.2.0/24

✓ Longest Prefix Match

Step 2: Next Hop

Next Hop

10.0.0.3

Step 3: Verify Destination

Subnet

192.168.2.0/24

Host Range

192.168.2.1 – 192.168.2.254

Destination

192.168.2.100

✓ Valid

Step 4: Routing Errors

Optimized VLSM Scheme

Example requirements

Department	Hosts Needed	Recommended Subnet
A	30	/26
B	25	/27
C	10	/28

Benefits

- Better utilization
 - Less wastage
 - Supports future growth
 - Efficient routing
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Final Answers Summary

Question	Final Answer
Q1	IP belongs to 192.168.10.64/26, gateway is incorrect, correct gateway should be in 192.168.10.64/26, usable hosts = 62.
Q2	Both devices belong to 2001:db8:0:0::/64; no prefix mismatch; corrected example addresses: 2001:db8::10/64, 2001:db8::20/64; available hosts = 2 ⁶⁴ .
Q3	Fixed subnetting wastes addresses; optimize using VLSM to allocate /26, /27, and /28 according to department sizes.
Q4	Same as Q2: both devices are on 2001:db8:0:0::/64; no addressing mismatch; available hosts = 2 ⁶⁴ .
Q5	Destination 192.168.2.100 matches 192.168.2.0/24; next hop is 10.0.0.3; if no route matches, the default route (10.0.0.1) is used.

Q2 is